

Day 33 - Lagrangian Examples

Me on my way to learn Lagrangian formalism



Me after learning the Lagrangian formalism



Interview Study - GenAI

- **Basma AlMahmood** is a PER PhD student at MSU.
- She is doing research on how students use generative AI in their (astro)physics classes.
- She is piloting an interview protocol to understand how students are using generative AI in their (astro)physics classes.

Physics Students AI Use Study



Announcements

- Homework 8 has been posted (Last HW; Due 17 Apr, "Late" 24 Apr)
 - Last Exercise 0: Reflect Learning Outcomes
- Final Project is posted
 - Video Presentations due 27 Apr
 - Computational Essay due 1 May
 - Rubric for both are posted
- No class (20 Apr - 24 Apr) - DC out of country
 - Make appointment for project help (clicker extra credit)

Reminder: The Lagrangian

The Lagrangian $\mathcal{L}$ is a function that summarizes the dynamics of the system:

$$\mathcal{L}(q, \dot{q}, t) = T - V$$

where:

- T is the **kinetic energy** of the system (depends on **gen. vel.**, $\dot{q}$),
- V is the **potential energy** of the system (depends on **gen. pos.**, q).

The equation of motion is recovered by applying the Euler-Lagrange equation to the Lagrangian (*minimizing the action integral*).

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) - \frac{\partial \mathcal{L}}{\partial q} = 0$$

Clicker Question 33-1

For a 1D SHO, the kinetic and potential energy are given by:

$$T = \frac{1}{2}m\dot{x}^2 \quad \text{and} \quad V = \frac{1}{2}kx^2$$

What are the derivatives of the Lagrangian $\mathcal{L} = T - V$ with respect to x and $\dot{x}$?

1. $\frac{\partial \mathcal{L}}{\partial x} = kx$ and $\frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x}$
2. $\frac{\partial \mathcal{L}}{\partial x} = -kx$ and $\frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x}$
3. $\frac{\partial \mathcal{L}}{\partial x} = kx$ and $\frac{\partial \mathcal{L}}{\partial \dot{x}} = -m\dot{x}$
4. $\frac{\partial \mathcal{L}}{\partial x} = -kx$ and $\frac{\partial \mathcal{L}}{\partial \dot{x}} = -m\dot{x}$
5. None of the above.

Clicker Question 33-2

For the plane pendulum, with $\mathcal{L}(x, \dot{x}, y, \dot{y}, t) = \frac{1}{2}m (\dot{x}^2 + \dot{y}^2) - mgy$

We found:

$$\frac{d}{dt}(m\dot{x}) = 0 \quad \text{and} \quad \ddot{y} = -g$$

Does that seem right?

1. Yes, it's fine.
2. Maybe, but I'm not sure I can tell you why.
3. No, I know this is wrong, but I'm not sure why.
4. No, this is definitely wrong and I can prove it!

Clicker Question 33-3

For the plane pendulum, we changed the Lagrangian from Cartesian coordinates to plane polar coordinates. In Cartesian, we found the Lagrangian depended on $y, \dot{x}, \dot{y}$. In polar, it only depended on ϕ and $\dot{\phi}$.

$$\mathcal{L}(x, y, \dot{y}) \longrightarrow \mathcal{L}(\phi, \dot{\phi})$$

What does that tell you about the dimensions of the system? The system is:

1. in 3D space, so it's 3D.
2. described by two spatial dimensions (x, y) , so it's 2D.
3. described by one spatial dimension (ϕ) , so it's 1D.

We chose our generalized coordinates poorly

We used the Lagrangian formalism to derive the equations of motion for a plane pendulum. We chose the x and y coordinates.

$$T(\dot{x}, \dot{y}) = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2), \quad V(y) = mgy \longrightarrow \mathcal{L} = T - V = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - mgy$$

This gave us the following derivatives for the Lagrangian:

$$\frac{\partial \mathcal{L}}{\partial x} = 0 \quad \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = \frac{d}{dt} (m\dot{x}) = 0$$

$$\frac{\partial \mathcal{L}}{\partial y} = -mg \quad \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{y}} \right) = -m\ddot{y}$$

We made a mistake by not including the constraint

We made a mistake by not including the constraint $x^2 + y^2 = L^2$ in our Lagrangian.

We can change variables to r and ϕ .

$$x = r \cos(\phi) \quad y = r \sin(\phi)$$

$$T(\dot{x}, \dot{y}) = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) = \frac{1}{2}m \left(r^2 \dot{\phi}^2 + 2r\dot{r}\dot{\phi} + \dot{r}^2 \right) = T(r, \dot{r}, \phi, \dot{\phi})$$

$$V(y) = mgy = mgr \sin(\phi) = V(r, \phi)$$

We made a mistake by not including the constraint

Now we include the constraint $r = L$, so that $\dot{r} = 0$.

$$T(\phi, \dot{\phi}) = \frac{1}{2}mL^2\dot{\phi}^2 \quad V(\phi) = mgL \cos(\phi)$$

$$\mathcal{L} = \frac{1}{2}mL^2\dot{\phi}^2 - mgL \cos(\phi)$$

Clicker Question 33-4

For the plane pendulum, we changed the Lagrangian from Cartesian coordinates to plane polar coordinates. In Cartesian, we found the Lagrangian depended on $y, \dot{x}, \dot{y}$. In polar, it only depended on ϕ and $\dot{\phi}$.

$$\mathcal{L}(x, y, \dot{y}) \longrightarrow \mathcal{L}(\phi, \dot{\phi})$$

What does that tell you about the dimensions of the system? The system is:

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Clicker Question 33-5

With $\mathcal{L} = \frac{1}{2}mL^2\dot{\phi}^2 - mgL\cos(\phi)$, we can find the equations of motion.

$$\frac{\partial \mathcal{L}}{\partial \phi} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \right) = 0$$

Which of the following equations of motion is correct?

1. $\ddot{\phi} = -\frac{g}{L}\sin(\phi)$
2. $\ddot{\phi} = -\frac{g}{L}\cos(\phi)$
3. $\ddot{\phi} = -\sqrt{\frac{g}{L}}\sin(\phi)$
4. $\ddot{\phi} = -\sqrt{\frac{g}{L}}\cos(\phi)$
5. None of these

Clicker Question 33-6

For the Atwood's machine, M is connected to m by a string of length l . Each mass has a length of string extended as measured from the center of the pulley (R) of y_1 and y_2 , respectively. The string wraps around half the pulley.

Which of the following represents the equation of constraint for the system?

1. $y_1 + y_2 = l - R\phi$
2. $y_1 - y_2 = l + R\phi$
3. $y_1 + y_2 = l - \pi R$
4. $y_1 - y_2 = l + \pi R$
5. None of these

Take the time derivative of the constraint equation. What do you notice?

Clicker Question 33-7

With a Lagrangian of the form $\mathcal{L} = \frac{1}{2}(M + m)\dot{y}_1^2 - (M - m)gy_1$, we can find the **generalized forces** and **generalized momenta**.

$$F_{y_1} = \frac{\partial \mathcal{L}}{\partial y_1} = -\frac{\partial V}{\partial y_1} \quad p_{y_1} = \frac{\partial \mathcal{L}}{\partial \dot{y}_1} = \frac{\partial T}{\partial \dot{y}_1}$$

What are F_{y_1} and p_{y_1} for the Atwood's machine?

1. $F_{y_1} = -mg$ and $p_{y_1} = m\dot{y}_1$
2. $F_{y_1} = -Mgy_1$ and $p_{y_1} = M\dot{y}_1$
3. $F_{y_1} = -(M - m)g$ and $p_{y_1} = (M + m)\dot{y}_1$
4. $F_{y_1} = -(M + m)g$ and $p_{y_1} = (M - m)\dot{y}_1$
5. None of these

Clicker Question 33-8

Now, we allow the pulley (mass, M_p) to rotate. The Lagrangian is given by:

$$\mathcal{L} = \frac{1}{2}(M + m)\dot{y}_1^2 + \frac{1}{2}I\dot{\phi}^2 - (M - m)gy_1$$

Where I is the moment of inertia of the pulley. What is the moment of inertia of the pulley?

1. $I = \frac{1}{2}M_pR^2$
2. $I = \frac{1}{3}M_pR^2$
3. $I = M_pR^2$
4. $I = \frac{1}{4}M_pR^2$
5. None of these

Clicker Question 33-9

The rope moves without slipping on the pulley. A rotation of $Rd\phi$ corresponds to a displacement of dy_1 for the first mass, M . What is the **new** equation of constraint for the system?

1. $y_1 + y_2 = l - R\phi$

2. $dy_1 = Rd\phi$

3. $y_1 = R\phi$

4. $\dot{y}_1 = R\dot{\phi}$

5. More than one of these